Aging reduces shared event structure at encoding, weakening memory reinstatement in the Posterior Medial Network

Supplementary Materials

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METHODS

Recall Specificity Scoring Methods

The television episode divided into meaningful transition and consisted of 37 possible events and 195 detail descriptions. Each event was annotated. Two independent raters independently segmented each participants' recall and labeled each segment according to the event it pertained to. Raters then scored each event for specificity by awarding gist and detail units. Gist units were awarded when a participant conveyed the general meaning of an event description. Detail units were awarded when the participant additionally recalled specific components (e.g., who, where, what). Detail scoring was nested within gist such that any detail unit necessarily implied the corresponding gist unit, whereas gist units could be awarded in the absence of detail. Below are the instructions provided to the raters:

Specificity Ratio Scoring Recall

1. Go to the recall schedule and check your assigned subject for the week.
2. Go to the recall_transcript folder on Box and open the assigned subject's transcript. Ensure that the audio (in the audio folder) matches the transcript. If not, edit the document before starting scoring procedure.
3. Segment the recall transcripts into "meaningful chunks."
4. Match the recall transcripts to the event it pertains to.
5. Open the scoring document spreadsheet.
6. Copy the template tab at the bottom. Replace the name with the participant number.
7. Complete the Subject column with the appropriate subject number.

8. Cross-reference the subject's transcript with the Detail Description column on the scoring sheet. You will enter 0 or 1 for each row for the following columns (Gist Unit and/or Detail Unit). Criteria for scores:
    a. Enter 1 for Gist Unit if the participant recalls only part of the information under the Detail Description (e.g., misses a character, fails to mention an integral part of the recall, etc.).
    b. Enter additional points for Detail Unit if the participant recalls additional part of the information completely and accurately.
    c. Leave the 0s if the participant fails to mention any of the information. Note: Yes, this is somewhat subjective but try your best!
9. For Recall Order column, mark the order in which the participant recalls the information.
10. Double check your work!
11. Repeat for the remaining assigned subjects.

Below is an example of the spreadsheet (Supplementary Table 1):

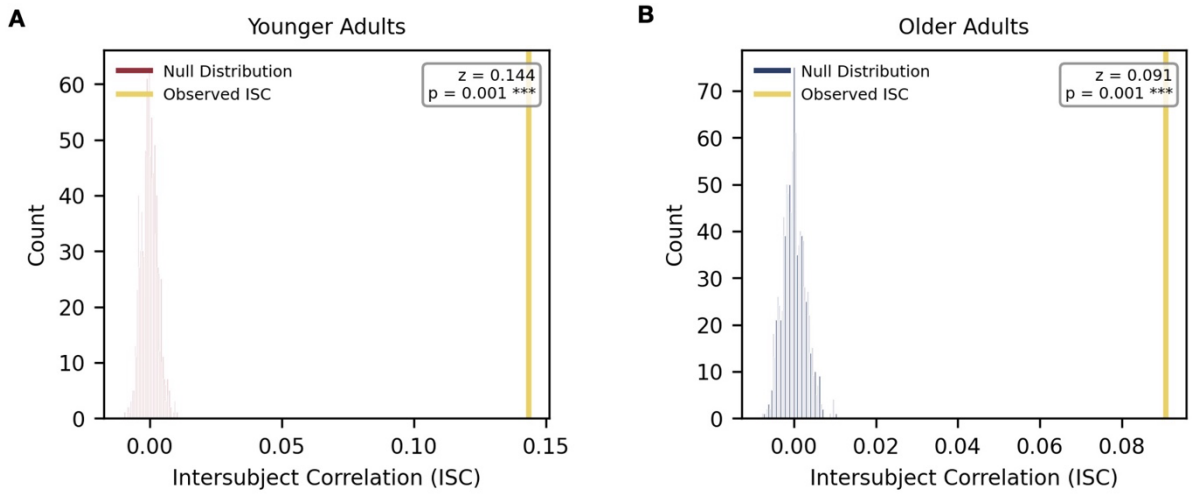
Supplementary Table 1. Recall Specificity Scoring Method Example.

ID	Recall	Detail Description	Event	Gist Unit	Detail Unit	Recall Order
30	Two men, one tall and one fat are walking down a street.	Jeff and Larry are walking down the street.	A1	1	0	1
30	-	Larry is carrying a paper bag.	A2	0	0	N/A
30	They run into a homeless man who begs for money.	Jeff and Larry encounter a homeless man who asks for change.	A3	1	1	2
…						
30			**Total**	2	1	

RESULTS

Intersubject Correlation Circular Time-Shift Permutation

To ensure that intersubject correlation reflected a shared reliable signal rather than noise, we conducted circular time-shift permutation tests (Nastase et al., 2019) with 1000 iterations separately for each age group. Both older (z = 0.091, p = .001) and younger (z = 0.144, p = .001) adults showed ISC significantly greater than zero (Supplementary Fig. 1), confirming reliable shared neural responses during encoding.



Supplementary Figure 1. (A) Intersubject Correlation Circular Time-Shift Permutation for Younger Adults and **(B)** Older Adults.